

SHREY PANWALA

☎ +91 87582 54494 ✉ shrey.panwala10@gmail.com 🔗 linkedin 🐙 github

Summary

AI/ML engineer and third-year Integrated B.Tech (CSE) + MBA student at Nirma University with hands-on experience building production-grade systems across computer vision, NLP/LLMs, RAG pipelines, and federated learning. Published author at **ACM MobiHoc '25, Houston, TX, USA**, two-time hackathon finalist, and IEEE General Secretary — proficient across the full ML pipeline from data engineering and model training to cloud deployment and real-time inference.

Education

Nirma University

Integrated B.Tech (Computer Science & Engineering) + MBA
– CGPA: 8.52 / 10

May 2028
Ahmedabad, India

Technical Skills

Languages: Python, C++, Java

AI/ML: Deep Learning, Computer Vision, NLP, Generative AI, LLMs, Federated Learning, Reinforcement Learning

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face Transformers, LangChain, FastAPI, Ultralytics, FAISS

Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Tableau, KNIME

Databases: MySQL, MongoDB, PostgreSQL

Tools: Git, Docker, AWS, Jupyter Notebook, Google Colab, Kaggle, VS Code, IntelliJ IDEA

Projects

Smart Parking & Traffic Intelligence System | *Computer Vision, XAI, FastAPI* | **Live Demo**

- Engineered a real-time traffic analytics platform using **YOLOv8** for multi-class vehicle detection and congestion estimation from live video streams; containerized the **FastAPI** backend with **Docker** and deployed on Vercel.
- Designed an XAI congestion pipeline with 3-tier severity classification (Low/Medium/High), SHAP-based explanations, and a React dashboard with heatmap overlays and automated parking scoring via 95th-percentile baselines.

AI News OS | *NLP, RAG, LLMs, Multi-Agent* | **ET Hackathon**

- Architected a production-ready **RAG** pipeline using **FAISS** vector indexing and **Hugging Face** sentence embeddings for semantic search over real-time news streams with context-aware retrieval.
- Orchestrated a 4-agent **LangChain** pipeline powered by **LLaMA 3** (via **Groq API**) for multilingual summarization and sentiment analysis; backed by FastAPI and PostgreSQL for low-latency data handling.

SmartShelf AI | *Computer Vision, ML Pipeline, GenAI* | **Top 10 Finalist – DA-IICT Hackathon 2026**

- Built a retail intelligence platform using **YOLOv8** detection and **ResNet-50** classification for automated shelf inventory tracking, real-time stock-out alerts, and continuous inference pipelines.
- Implemented a demand forecasting ensemble (**Prophet + LightGBM**) for time-series inventory prediction and a multilingual LLM assistant for natural-language restocking recommendations and retail decision support.

Research & Publication

FedQ-Fraud: A Quantum-Enforced Federated Learning Framework for Financial Fraud Detection

2025

- Published at **ACM MobiHoc '25, Houston, TX, USA** – built a quantum-enforced FL framework using a 3-layer GRU with MOON, achieving **97.47%** fraud detection accuracy across 4 non-IID clients; secured model transmission via BBM92 QKD + Bonawitz-style Secure Aggregation against gradient poisoning and data heterogeneity.

Achievements

- **Top 10 Finalist** – DA-IICT Hackathon *Tic-Tech-Toe 2026* for SmartShelf AI retail intelligence system.
- **Ranked 6th** – HackNUthon for *ReeveSoils*, an ML-driven solution for soil analysis and crop yield prediction.
- **ML + Deep Learning + NLP Specializations (Coursera – DeepLearning.AI)** – Supervised ML, Advanced Algorithms, Neural Networks, CNNs, RNNs, NLP Classification, Probabilistic Models & Attention Models.

Positions & Responsibilities

IEEE ITNU – Intelligent Transportation Systems Society

General Secretary

- Organized technical workshops on intelligent transportation systems, coordinated external speakers, managed event logistics, and drove student participation across competitions and seminars.